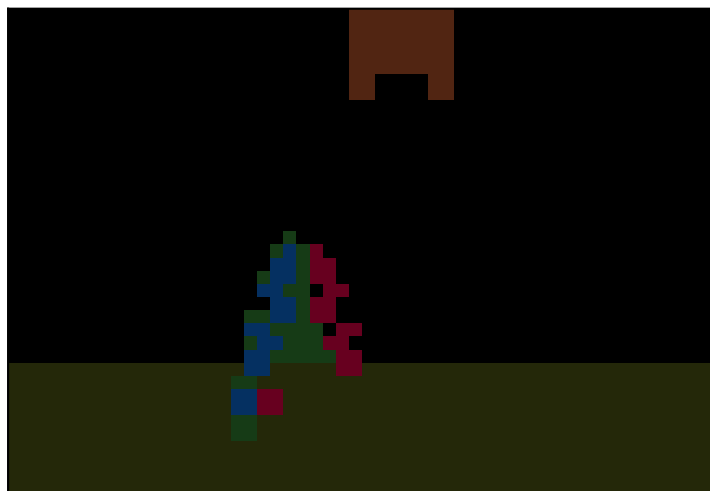


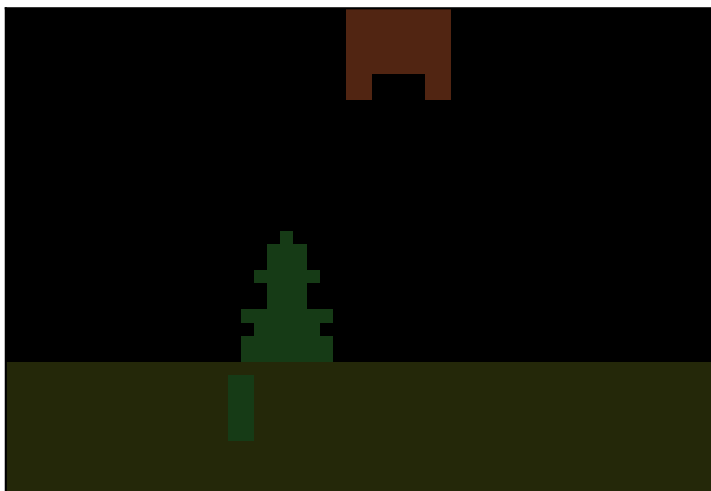
(a) real Space Invaders, 35 s (game just begun)



(b) $\partial \text{screen} / \partial \text{RIGHT}$ — sampler
(identical for all 3 variants, $\max|\Delta|=0$)



(c) $\partial \text{screen} / \partial \text{RIGHT}$ — naive index $\equiv 0$ (vanishes)



(d) inverse: $\partial(\text{move cannon right}) / \partial \text{joystick}$

